Samih Eisa Suliman Abdalla

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Samih Eisa - Google Scholar

Summary

I am a postdoctoral researcher at INESC-ID, University of Lisbon, with expertise in distributed systems, Internet of Things (IoT), and digital twin technologies for health applications. I have contributed to multiple national and international research projects and have a track record in mentoring students and publishing in peer-reviewed venues. My current research focuses on enabling continuous update and validation of medical digital twins to support clinical decision making.

EDUCATION

PhD	**Computer Science | University Minho, Aveiro, Porto** Behavioral Modeling for Ambient Assisted Living	April 2018
MSc	**Computer Science | University Malaya** Data Communication and Computer Networks	August 2009
BSc	**Computer Science | Sudan University** Computer and Information Systems	June 2003

EMPLOYMENT

2024 – current
PostDoc Researcher | INESC-ID, University of Lisbon, Portugal

Contributes to the BLOCKCHAIN.PT project, “Descentralizar Portugal com Blockchain”, national initiative, supporting research and technological development in decentralized systems. Actively involved in project deliverables, development activities, dissemination, and co-supervision of Master’s students. Conducted research and scientific activities resulted in 8 publications and preprints so far.

2020 – 2022
PostDoc Researcher | INESC-ID, University of Lisbon, Portugal

Conducted research on device location certification for IoT systems, resulting in multiple peer-reviewed publications (7 publications in reputed conferences). Contributed to project development, authored deliverables, and co-managed research activities while supervising MSc students.

2018 - 2020
PostDoc Researcher/Senior Development Technician | Centro de Computação Gráfica, Guimarães, Portugal

Worked within the Computer Vision, Interaction and Graphics group on applied R&D projects in computer vision, augmented reality, and Industry 4.0, in collaboration with industry partners.

2016 - 2018
Researcher / Software Developer | Centro Algoritmi, University of Minho, Guimarães, Portugal

Developed communication mechanisms for smart autonomous systems in collaboration with Bosch-UMinho (SAMU project), contributing to system design, development, and deliverables.

2011 – 2016
Researcher/Software Developer | Centro de Computação Gráfica, Guimarães, Portugal

As part of the Urban and Mobile Computing (UMC) research group, contributed to multiple R&D projects in mobile computing and smart environments.

2010 – 2011
Software Engineer | Scientific Computing (SciCom) Ltd Company, Khartoum, Sudan

Handled various software development tasks.

2009 – 2010
Research Assistant | University of Malaya, Kuala Lumpur, Malaysia

Conducted research on resource allocation in Computer Networks using Reinforcement Learning.

2007 – 2008
Lecturer | Cosmopoint International University College, Kuala Lumpur, Malaysia

Taught programming and database subjects to undergraduate students. Developed and updated course curriculums to ensure relevancy and applicability.

2006 – 2007
Teaching Assistant | University Technology Petronas, Perak, Malaysia

Assisted with teaching of programming subjects, aiding in lab sessions and grading assignments. Helped the students in understanding complex concepts, improving class performance.

2003 – 2005
Teaching Assistant | Sudan University, Khartoum, Sudan

Taught programming, database, web development and distributed systems, effectively communicating complex information to students. Mentored undergraduate students in their final-year projects.

RESEARCH INTEREST

- Digital Twins for Health.
- Personalized and Adaptive Modeling.
- Uncertainty Quantification (VVUQ).
- Indoor Positioning and Tracking.
- Remote Patient Monitoring.

ACADEMIC ACTIVITY

- **Student Supervision**
    - Co-supervision of MSc students (16+ completed)

    - Support in dissertation development and scientific publication

- **Teaching Experience**
    - Programming, databases, distributed systems, and web technologies
    - Experience across multiple institutions (Malaysia and Sudan)

PROFESSIONAL SERVICE

- **Member** of Distributed, Parallel, and Secure Systems Group, INESC-ID, Lisbon, Portugal, 2020-current
- **Member** of Computer Vision Group, Centro de Computação Gráfica, Guimarães, Portugal, 2018-2020
- **Member** of Ubiquitous Systems research group, (Ubicomp@UMinho), Guimarães, Portugal, 2012-2018

- **Evaluation Committee Member** of the EvAAL initiative for Indoor Localization Competition, Madrid, Spain, 2013

- **Peer-Reviewed for**:
    - IEEE Journal of Biomedical and Health Informatics
    - IEEE NCA
    - IUCC-CSS
    - IPIN
    - IFIP-SEC, SAC, NOMS/AIMLOps

COMMUNITY AND OUTREACH

Contributed to education and outreach through:

- Development of teaching materials (University of Malaya)
- Programming competitions for school and university students (Sudan and Malaysia)
- Promotion of computing education and skills development

PROJECTS

Participation in several national and international R&D projects span IoT, blockchain, Industry 4.0, smart mobility, and Ambient Assisted Living.

- BLOCKCHAIN.PT **(2024):** Decentralize Portugal with Blockchain **| Researcher**
- SureThing **(2022):** Device Location Certification for the Internet of Things **| Researcher**
- **Q-better (2020):** Visual Analytics Training for developers **|Instructor / Materials Creator**
- **Intelligent4DMoulds (2020):** AR-based Mould Design system **| Researcher/ Developer**
- **UH4SP (2020):** Unified-Hub for Smart Plants**| Researcher / Software Developer**
- **SAMU (2018):** Smart Autonomous Mobile Units**| Researcher / Software Developer**
- **ONLY (2016):** Smart Home Gateway for domotics **|Software Developer**
- MOSAIC 2B **(2016):** Mobile Technologies for Business (EU project) **| Software Developer**
- **AAL4ALL (2014):** Ecosystem for Ambient Assisted Living **| Researcher / Software Developer**
- **TriallMed (2014):** Middleware for remote e-Health**| Researcher / Software Developer**.

KEY STRENGTHS

- Interdisciplinary research across IoT, AI, and healthcare
- Strong collaboration with academia and industry
- Experience in mentoring and team development
- Focus on real-world impact and translational research

SUMMARY OF PHD WORK

Conducted research on the analysis of in-home mobility behaviour of the living alone elderly people using sensor data, contributing valuable insights to the field of Ambient Assisted Living by developing an unsupervised probabilistic approach to understand daily mobility routines of the elderly and detect unusual behaviours, enhancing our ability to provide supportive care for the elderly.

SCIENTIFIC OUTPUT

Author and co-author of multiple peer-reviewed publications in international conferences and journals, including IEEE venues in areas such as:

- IoT security and location certification
- Blockchain-based systems
- Indoor positioning and tracking
- Ambient Assisted Living

Selected works include contributions to IEEE NCA, EDCC, VTC, and Sensors journal.

SELECTED PUBLICATIONS

Nunes, E., **Samih Eisa**, Pardal, M.L., & Calha, M. (2025). **CC2C: Confidential Channel-to-Channel Data Exchange in a Permissioned Blockchain**. 2025 23rd International Symposium on Network Computing and Applications (NCA), 300-303.

Faria, F, **Samih Eisa**, Matos, D.R., & Pardal, M.L. (2025). **EvoChain: A Recovery Approach for Permissioned Blockchain Applications.** 2025 23rd International Symposium on Network Computing and Applications (NCA), 276-285.

Avelar, A., **Samih Eisa**, Remédios, O., & Pardal, M.L. (2025). **ChainGuard: Verified Data Intake for a Track & Trace Blockchain.** 2025 20th European Dependable Computing Conference Companion Proceedings (EDCC-C), 168-173.

Preto, L., **Samih Eisa**, Remédios, O., Matos, D.R., & Pardal, M.L. (2025). **Dependable Food Traceability Data Through Blockchain and Database Integration.** 2025 20th European Dependable Computing Conference Companion Proceedings (EDCC-C), 163-167.

Gomes, B., **Samih Eisa,** Matos, D.R., & Pardal, M.L. (2024). **Decentralized Storage And Self-Sovereign Identity For Document-Based Claims.** ArXiv, abs/2411.16987.

Faria, F., **Samih Eisa**, Matos, D.R., & Pardal, M.L. (2024). **EvoChain: a Recovery Approach for Permissioned Blockchain Applications**. ArXiv, abs/2411.16976.

F. Pollicino, **Samih Eisa**, P. Rosa, M. L. Pardal and M. Marchetti, "**Decentralized position detection for moving vehicles**," 2023 IEEE 97th Vehicular Technology Conference (VTC2023-Spring), Florence, Italy, pages 1-6, 2023.

Lucas H. Vicente, **Samih Eisa**, and Miguel L. Pardal. **Locaas: Location-certification-as-a-service**. In 21st IEEE International Symposium on Network Computing and Applications, NCA 2022, Boston, MA, USA, December 14-16, 2022, pages 165–172. IEEE, 2022.

João Tiago, **Samih Eisa**, and Miguel L. Pardal. **Surespace: orchestrating beacons and witnesses to certify device location**. In SAC ’22: The 37th ACM/SIGAPP Symposium on Applied Computing, Virtual Event, April 25 - 29, 2022, pages 164–173. ACM, 2022.

Ricardo Grade, **Samih Eisa**, and Miguel L. Pardal. **Bluetooth peer-to-peer location certification with a gamified mobile application**. In 21st IEEE International Symposium on Network Computing and Applications, NCA 2022, Boston, MA, USA, December 14-16, 2022, pages 55–62. IEEE, 2022.

Rafael Figueiredo, **Samih Eisa**, and Miguel L. Pardal. **Surerepute: Reputation system for crowdsourced location witnesses**. In 21st IEEE International Symposium on Network Computing and Applications, NCA 2022, Boston, MA, USA, December 14-16, 2022, pages 261–268. IEEE, 2022.

Rui Claro, **Samih Eisa**, and Miguel L. Pardal. **Lisbon hotspots: Wi-fi access point dataset for time-bound location proofs**. CoRR, volume abs/2208.04741, 2022.

Miguel C. Francisco, **Samih Eisa**, and Miguel L. Pardal. **Secure protocol buffers for Bluetooth low-energy communication with wearable devices**. In 20th IEEE International Symposium on Network Computing and Applications, NCA 2021, Boston, MA, USA, November 23-26, 2021, pages 1–8. IEEE, 2021.

Samih Eisa and Adriano J. C. Moreira. **A behaviour monitoring system (BMS) for ambient assisted living**. Sensors, volume 17, page 1946, 2017.

Samih Eisa, João Peixoto, Filipe Meneses, and Adriano J. C. Moreira. **Removing useless aps and fingerprints from wifi indoor positioning radio maps**. In International Conference on Indoor Positioning and Indoor Navigation, IPIN 2013, Montbeliard, France, October 28-31, 2013, pages 1–7. IEEE, 2013.

Samih Eisa and Adriano J. C. Moreira. **Requirements and metrics for location and tracking for ambient assisted living**. In 2012 International Conference on Indoor Positioning and Indoor Navigation, IPIN 2012, Sydney, Australia, November 13-15, 2012, pages 1–7. IEEE, 2012.